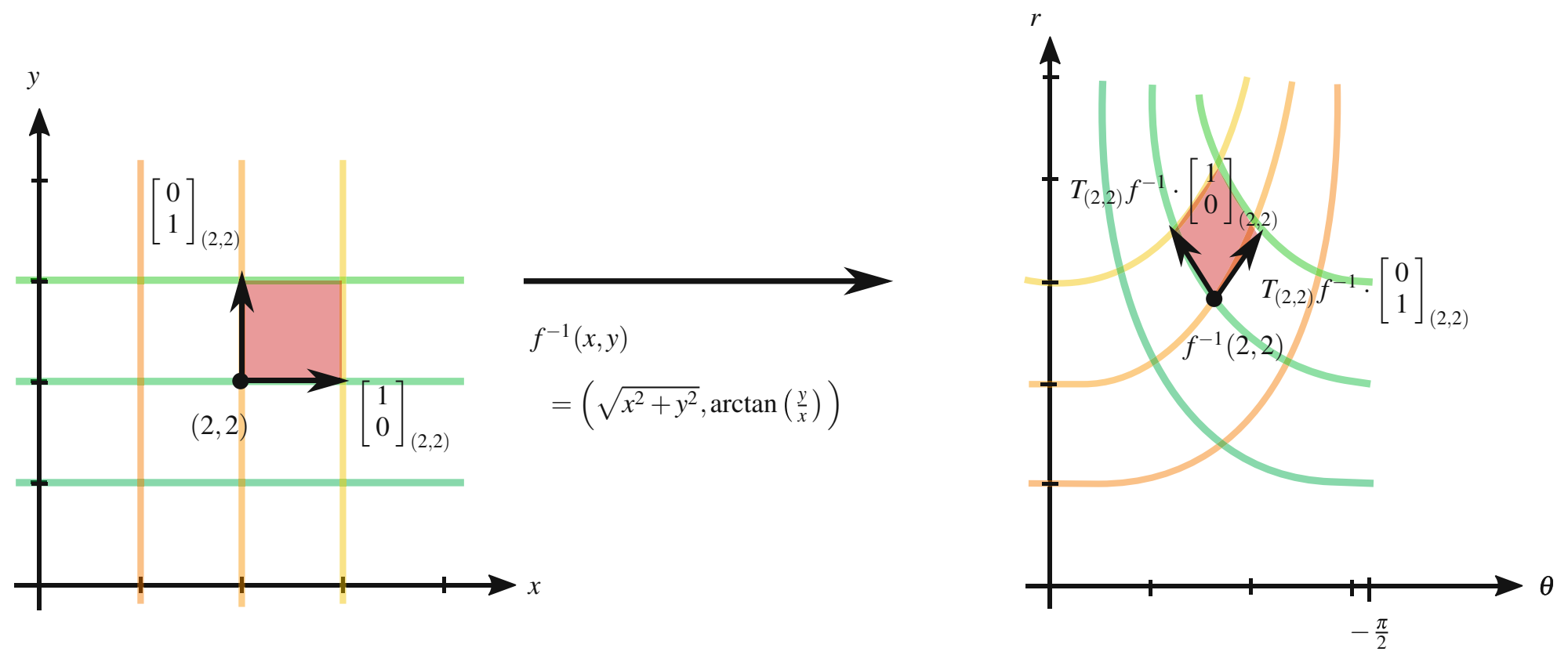

y
x
$\begin{bmatrix}0\\1\end{bmatrix}_{(2,2)}$
(2,2)
$\begin{bmatrix}1\\0\end{bmatrix}_{(2,2)}$
$f^{-1}(x,y)$
$= \left(\sqrt{x^2+y^2}, \arctan\left(\frac{y}{x}\right)\right)$
r
θ
$T_{(2,2)}f^{-1} \cdot \begin{bmatrix}1\\0\end{bmatrix}_{(2,2)}$
$T_{(2,2)}f^{-1} \cdot \begin{bmatrix}0\\1\end{bmatrix}_{(2,2)}$
$f^{-1}(2,2)$
$-\frac{\pi}{2}$